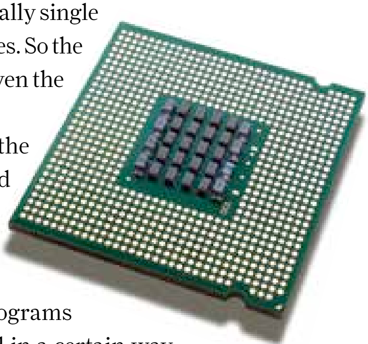
Technology

1. Core

The word core means different things for both processor manufacturers. Intel has a few actual cores and then you have HyperThreading which emulates two threads (think of a thread as a processing queue) per core. So the operating system sees two cores per core ... confusing isn't it? Basically just multiply the number of cores by 2 if you see the HT (HyperThreading) tag on the model. This, however, doesn't mean that there are actually that many physical cores. AMD has a similar architecture where the processor is divided into certain modules and each module though physically single is capable of performing the same as 2 cores. So the word core has lost a bit of significance given the hoopla generated by either company.



A common misconception is that if the processor is rated at say 2 GHz then a quad core processor is equal to 8 GHz, this is definitely not how it works. Performance does not scale in a linear fashion. What matters even more is the way programs are coded, they need to be programmed in a certain way in order to utilise the power of more than one core so don't expect any program to speed up miraculously when you get more cores.

Another concern is how many cores a user needs. Most programs released after Windows 7 emerged are optimised for multiple cores, however, when benchmarked it was observed that at most 2 cores were utilised. Video games share a similar fate, few games are programmed to handle 4 cores while the majority work fine with 2 cores. Multimedia Design software on the other hand use as many cores as it can get its hands on. So for gaming 4 cores without hyperthreading is the best choice, video editing and 3D modelling do well with the enthusiast processors and casual home / office users need no more than 1 / 2 core(s).

2. Sockets

We've covered sockets in good detail in the next section on motherboards, so we'll just have an overview here. Each processor generation came with it's own socket, sometimes manufacturers like to use the same socket as the previous generation but of late what we have seen is a change in socket types every other year.